

History of Integrated Motherboard Chipset

The Motherboard industry has, until now, been dominated by chipsets such as Intel, VIA and Ali which provide integrated audio and video on the motherboard itself. These chipsets, however, are in no way as powerful as the nForce chipsets are going to be. The Intel range of integrated chipsets includes the OpenGL-compliant Intel 810, which uses the Intel 754 graphics card, one of the oldest 3D accelerator cards with OpenGL support. While the chipset's performance seemed pretty good for the time, competition slowly began to grow in the form of early nVidia and ATI graphics cards. VIA also has and has had chipsets with integrated sound and graphics. Similarly, the i810 gives integrated audio and video, which is a cheaper option for the home user. Ali, with its sound and audio integrated Aladdin chipsets, has been offering AC 97 audio to its users. There have been some motherboards with inbuilt modems and network adapters too, but never before have all these high-end components come together on one main motherboard.

nVidia ventured into the graphics card business with the Riva 128, slowly moved into GeForce based chipsets, and are now creating motherboard chipsets with high-end performance. Having recently worked in tandem with Microsoft on the Xbox, nVidia decided to make use of experts and resources and enter the motherboard market.

give you an optimal performance machine? To start with, the IGP plays the central role of communicating with the processor, the MCP and the 128-bit DDR memory. Here on though, the integrated graphics processor is of no use. The Twin-Bank Memory Architecture then comes into action, creating a huge bandwidth gap between the 128-bit DDR memory and the main IGP itself. DASP works side by side with the main processor of your computer, by determining future patterns. AMD's own Hypertransport, placed between the two bridges on the nForce chipset, provides a huge bandwidth of 800 MBps with the help of the dual 200 MHz DDR bus. Down south in the MCP, the APU delivers 5.1 Dolby Digital Sound to the speakers. The MCP also gives you a dual ATA100 hard disk connectivity, six USB connectors along with two USB hubs for connectivity to external drives, and an inbuilt 10/100 MBps Ethernet card with an integrated 56 Kbps modem. Streamthru technology in the MCP ensures streaming audio and video and also optimal network performance through the integrated network card.

What this means...

nVidia's entrance into the motherboard chipset industry implies direct competition for the companies already ruling the roost. With each of these companies specialising in one particular area, nVidia's most recent move can be seen as an attempt to take over all these areas and become an all rounder. Some of the companies that nVidia is taking on head to head are motherboard chipset companies such as Intel and VIA. Competition is also evident for graphics processor companies such as ATI, Matrox, etc, which have plans to give out high-end graphics processors integrated into motherboards like never before. Intel and VIA motherboards have never had the capability to deliver such high-end performance ever. The performance of the Geforce 2 MX like core in the nForce IGP cannot be matched by the VIA and Intel graphics processors.

Creative has been ruling the sound industry so far by providing home users with 5.1 surround sound using the Creative SBLive soundcard. But a change in scenery is evident as home users can now opt for a motherboard with a built-in 5.1 surround soundcard. nVidia claims

that their chipset-based motherboards will procure a better level of performance from existing AMD processors. Most of these technologies aren't available all in one place on the existing motherboards.

As far as the consumers are concerned, instead of having to live with the pathetic performance that most integrated motherboards give you, you can now avail of the hard-hitting performance of an nForce-based motherboard at a relatively reasonable price, with variants that offer you fewer features at a lower price. You will also have the flexibility to choose whether or not to upgrade to top-end graphics card (maybe the new GeForce3) and soundcard (the Creative SBLive! Platinum?) owing to the availability of slots for add-on cards. You also probably won't need to worry about buying a modem or LAN card anymore, because the integrated devices should be sufficient for most Windows users.

Those of you intent on buying a new branded PC, such as a Compaq or HP, might want to purchase one that offers an AMD-nForce combination. This is, of course, assuming that these manufacturers are sensible enough to use the nForce for their high-end systems at a lower price.

Our prediction

While the price tag of the nForce chipset is currently unknown, the cost of motherboards based on this chipset are likely to be anything between \$250 to \$350 (approximately Rs 11,500 to Rs 16,100), which is pretty decent for a motherboard packed with high performance components.

So what is the nForce really going to do to the computer world? The nForce could be adopted by system builders for creating performance machines aimed at price conscious users who are willing to pay a little extra for performance gains. If the prices drop further, as is quite likely over a period of time, the popularity of the nForce-based motherboards is bound to increase as compared to the i810 or i815 unit with its skimpy little graphics processor.

The first set of nForce-based motherboards should be out in a couple of months from now. nVidia's done it with the graphics card processor industry and now it's at it again, this time aiming for the bigger picture of computing. ■



An nForce reference board